Emre Kerman

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Summary

Computer Engineer from İzmir and recent Dokuz Eylül University graduate with experience in software development, machine learning, embedded systems, and full-stack web development. Experienced in building IoT and computer vision systems, developing frontend and backend applications, and working with Linux and open-source technologies.

Education

Dokuz Eylül University — Izmir, Turkey
Bachelor of Science in Computer Engineering — *2021 – 2025*

Yunus Emre Anadolu Lisesi — Izmir, Turkey
High School Diploma — *2018 – 2021*

Experience

Software Engineering Intern — June 2025 – Sept. 2025
Indas Teknoloji Ticaret AŞ. — *Izmir, Turkey*

- Engineered a smart parking platform using ESP32-CAM nodes and OCR (PaddleOCR/Tesseract) for automated vehicle entry tracking, billing, and balance management via Next.js REST APIs.
- Built an edge-first PPE and human detection system on ESP32-CAM using quantized FOMO models, C++/Arduino, and Edge Impulse, achieving 120–250ms inference latency, 18 FPS live streaming, and 5–6 FPS on-device inference through dual-buffering and PSRAM optimization.
- Developed web frontends using Next.js and React for real-time bounding box overlays, detection logs, and balance dashboards.

Software Engineering Intern — Nov. 2025 – Dec. 2025
Dokuz Eylül University Computer Engineering Dept. — *Izmir, Turkey*

- Developed "Restock", a self-checkout platform integrating physical store kiosks with mobile devices for real-time session handover.
- Implemented atomic DB transactions using Drizzle ORM and SQLite alongside custom React hooks for barcode scanning.
- Configured role-based auth (JWT) via Next.js Middleware and built administrative monitoring dashboards.

Projects

Early Frost Detection | *Go, Python, C, TensorFlow, Keras, NumPy, ESP-32* — 2024 – 2025

- Developed a machine learning-based IoT solution to predict ice formation on refrigerator cooling fins for an Arçelik use case.
- Designed and assembled a hardware refrigerator prototype, integrating ESP32 microcontrollers and DHT22 sensors to collect environmental data over Wi-Fi.
- Performed data preprocessing, feature engineering, and machine learning model training on the collected data.
- Built a real-time web server and front-end monitoring platform using Go.

Hand Gesture Detection | *Python, TensorFlow, Keras, NumPy, OpenCV* — 2024

- Constructed two computer vision models (a lightweight CNN and a ResNet-like architecture) to classify gestures via webcam.
- Curated and preprocessed a custom hand-gesture dataset for model training.
- Implemented an efficient training pipeline utilizing Model Checkpoints to prevent data loss.
- Integrated live webcam input with UX enhancements for real-time inference.

Game Platform & Storefront | *C#, .NET Core, PostgreSQL* — 2023

- Designed and implemented the complete database architecture for a digital game distribution platform.
- Developed backend application logic to support user authentication, game publishing, and purchasing.

Technical Skills

Languages: TypeScript, JavaScript, Python, C/C++, Go, C#, SQL, Rust
Web: Next.js, React, React Native, Astro, Tailwind CSS, REST APIs
Backend & Databases: PostgreSQL, SQLite, Drizzle ORM, .NET Core, OAuth/JWT, Swagger/OpenAPI
AI, Vision & Edge: TensorFlow, Keras, TFLite, OpenCV, Edge Impulse, ESP32/ESP32-CAM, PaddleOCR
Systems & DevOps: Linux, Docker, Kubernetes, Git, Nix/NixOS
Spoken Languages: Turkish (Native), English (C1), German (A1)